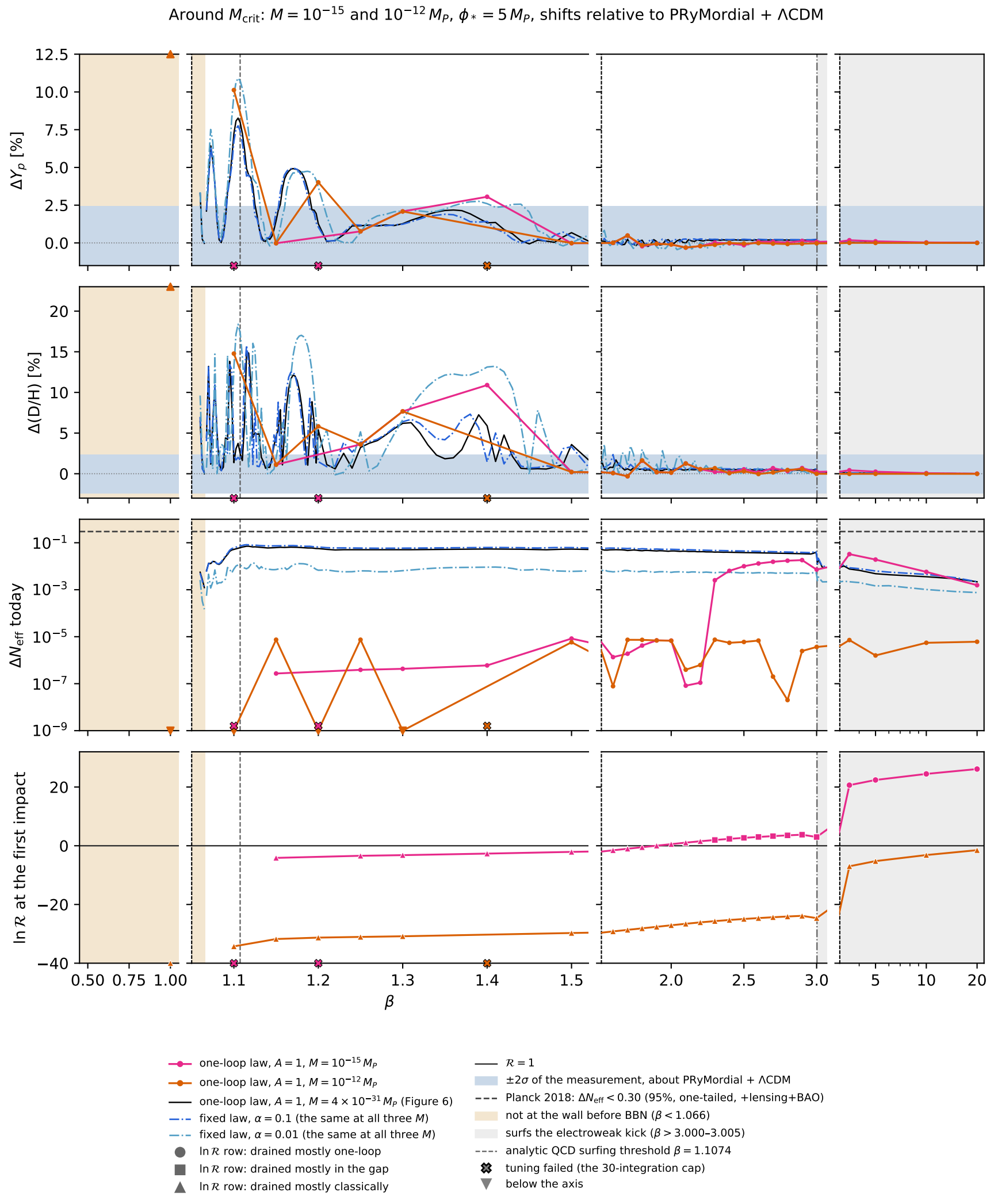

Around $M_{\rm crit}$: $M = 10^{-15}$ and $10^{-12}\,M_P$, $\phi_* = 5\,M_P$, shifts relative to PRyMordial + ΛCDM
ΔY_p [%]
Δ(D/H) [%]
$\Delta N_{\rm eff}$ today
ln $\mathcal{R}$ at the first impact
β
one-loop law, $A = 1$, $M = 10^{-15}\,M_P$
one-loop law, $A = 1$, $M = 10^{-12}\,M_P$
one-loop law, $A = 1$, $M = 4 \times 10^{-31}\,M_P$ (Figure 6)
fixed law, $\alpha = 0.1$ (the same at all three M)
fixed law, $\alpha = 0.01$ (the same at all three M)
ln $\mathcal{R}$ row: drained mostly one-loop
ln $\mathcal{R}$ row: drained mostly in the gap
ln $\mathcal{R}$ row: drained mostly classically
$\mathcal{R} = 1$
±2σ of the measurement, about PRyMordial + ΛCDM
Planck 2018: $\Delta N_{\rm eff} < 0.30$ (95%, one-tailed, +lensing+BAO)
not at the wall before BBN ($\beta < 1.066$)
surfs the electroweak kick ($\beta > 3.000$–3.005)
analytic QCD surfing threshold $\beta = 1.1074$
tuning failed (the 30-integration cap)
below the axis